

Predictive Boundary Establishment Module

Module: PBEM — Predictive Boundary Establishment Module

Parent Standard: POPS — Prediction Object & Purpose Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Predictive Boundary Establishment

Primary Governed Object: Predictive Boundary

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Predictive Boundary Establishment Module (PBEM) governs the establishment, specification, differentiation, coherence, and continuity of the boundary within which an identified Prediction Object, defined Predictive Purpose, specified Prediction Target, and defined Prediction Horizon together constitute the governed predictive proposition.

PBEM establishes what is included within and excluded from that proposition before the prediction proceeds into methodology and subsequent PGA™ governance.

The Predictive Boundary prevents related objects, outcomes, populations, conditions, locations, periods, variables, or interpretations from being silently incorporated into the original prediction.

2. Operational Role

PBEM is the fifth and final operational module within POPS.

It receives:

Identified Prediction Object + Defined Predictive Purpose + Specified Prediction Target + Defined Prediction Horizon

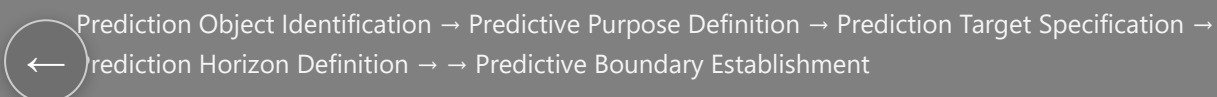
and establishes the boundary of the predictive proposition.

Its principal transition is:

Defined Predictive Frame Components → Established Predictive Boundary → Governed Prediction Object & Purpose

PBEM completes the POPS governance sequence and provides the governed predictive frame required by PMES — Prediction Methodology Standard .

3. Internal Module Flow



Current Position: Predictive Boundary Establishment

The completed POPS transition is:

POIM → PPDM → PTSM → PHDM → PBEM → Governed Prediction Object & Purpose

4. Operational Space

PBEM operates within the Predictive Boundary Establishment space.

It governs:

predictive inclusion; predictive exclusion; boundary dimensions; boundary specificity; boundary coherence; boundary ambiguity; boundary overlap; boundary dependencies; boundary conditions; boundary change; boundary expansion and contraction; boundary continuity.

PBEM governs the limits of the predictive proposition.

It does not govern the limits of Operational Reality itself.

5. Operational Scope

A Predictive Boundary may include or exclude dimensions such as:

objects; target states; populations; geographic areas; systems; environments; organizational units; markets; operational conditions; variables; event classes; temporal conditions; contextual conditions; other materially relevant predictive limits.

Not every prediction requires every boundary dimension.

PBEM governs those dimensions necessary to prevent material ambiguity or inappropriate expansion of predictive meaning.

6. Predictive Boundary Establishment

PBEM must establish where the governed predictive proposition begins and ends.

The boundary must be sufficiently clear to determine whether a later interpretation, outcome, condition, or use remains inside the original predictive proposition.

The governing question is:

What is legitimately included within this prediction, and what remains outside it?

A prediction whose boundaries remain materially undefined may be capable of producing an output while remaining methodologically ambiguous.

7. Boundary Inclusion

PBEM must identify elements whose presence materially defines the predictive proposition.

An inclusion may concern a population, geographic region, system, condition, event class, target interpretation, operational context, or another relevant dimension.

Included elements become part of the predictive frame to which subsequent methodology, basis, assumptions, probability, uncertainty, and feasibility must correspond.

Related Element ≠ Included Predictive Element

8. Boundary Exclusion

Material exclusions must remain identifiable where their omission affects interpretation of the prediction.

Exclusion prevents adjacent or related conditions from later being treated as though they were part of the original predictive proposition.

An exclusion does not establish that the excluded element is irrelevant in every context.

It establishes that it does not belong to the governed predictive proposition unless subsequently governed through an explicit change.

9. Boundary Dimensions

A Predictive Boundary may contain multiple dimensions.

These may include:

Object Boundary

Population Boundary

Geographic Boundary

System Boundary

Operational Boundary

Contextual Boundary

Conditional Boundary

Temporal Boundary

Target Boundary

The relevant combination depends upon the prediction.

Multiple boundary dimensions may coexist without becoming separate Prediction Objects.

10. Boundary Specificity

The Predictive Boundary must be sufficiently specific to preserve predictive meaning.

An excessively broad boundary may allow materially different outcomes to be interpreted as satisfying the same prediction.

An excessively narrow boundary may introduce artificial constraints unsupported by the Predictive Purpose.

PBEM therefore governs sufficient boundary specificity rather than maximal restriction.

Boundary Specificity $\neq$ Maximum Boundary Narrowness

11. Boundary Ambiguity

PBEM must identify ambiguity capable of materially changing what the prediction is understood to cover.

Material ambiguity may arise from:

unclear geographic scope; undefined population; unspecified system coverage; unclear contextual limits; ambiguous event inclusion; unresolved conditional boundaries; conflicting interpretations; implicit exclusions; inconsistent scope across predictive components.

A materially ambiguous boundary must not be treated as fully established.

12. Boundary Coherence

The Predictive Boundary must remain coherent with:

the Prediction Object; Predictive Purpose; Prediction Target; Prediction Horizon.

A boundary must not silently redefine any of these preceding governed elements.

Where boundary establishment reveals incompatibility, the affected POPS component must be clarified rather than forcing an incoherent predictive frame forward.

Boundary Establishment $\neq$ Permission to Redefine the Prediction

13. Boundary Overlap

Different predictions may possess overlapping boundaries.

Overlap does not make them the same prediction.

PBEM must preserve differentiation where predictive propositions share geography, population, systems, conditions, objects, or time periods but differ materially in object, purpose, target, horizon, or other boundary dimensions.

Boundary Overlap $\neq$ Prediction Equivalence

14. Boundary Dependencies

Some Predictive Boundaries depend upon external conditions or classifications.

Where a boundary depends upon another governed object, state, system, jurisdiction, population definition, or operational condition, that dependency must remain identifiable where material.

PBEM does not take governance ownership of the external dependency.

It governs its role in defining the predictive proposition.

15. Conditional Boundaries

A Predictive Boundary may apply only while specified conditions remain true.

Where conditionality materially determines whether something remains inside the predictive proposition, that condition must be identifiable.

Detailed governance of assumptions belongs to PAS — Prediction Assumption Standard .

PBEM establishes the boundary relationship without taking ownership of assumption governance.

Boundary Condition $\neq$ Predictive Assumption Governance

16. Geographic Boundaries

Where location materially affects predictive meaning, PBEM must establish the applicable geographic scope.

A prediction concerning a city, region, country, facility, market, network, or physical area must not automatically be generalized beyond its governed geographic boundary.

Geographic Proximity $\neq$ Predictive Inclusion

17. Population Boundaries

Where a prediction concerns people, organizations, systems, agents, assets, or another defined population, PBEM must establish which population is included.

A prediction derived from or concerning one population must not automatically be represented as applying to another.

Population representation and collective predictor composition are governed elsewhere where applicable.

Observed Population $\neq$ Automatically Governed Population

18. System and Operational Boundaries

Where predictions concern technical, organizational, AI, autonomous, infrastructure, or operational systems, PBEM must identify the system or operational scope to which the predictive proposition applies. Prediction concerning one subsystem, deployment environment, operational state, or organizational unit must not automatically be generalized to the entire system or organization.

Subsystem Prediction $\neq$ Whole-System Prediction

19. Predictive Boundary vs Operational Reality Boundary

The Predictive Boundary governs the scope of the predictive proposition.

It does not establish the boundaries of Operational Reality.

Operational Reality remains governed by the applicable reality-governance architecture, including ORA™ where relevant.

Predictive Boundary $\neq$ Operational Reality Boundary

PGA™ may predict something about Operational Reality without acquiring governance ownership of Operational Reality itself.

20. Boundary Expansion

A Predictive Boundary may legitimately expand after the original predictive frame has been established.

A material expansion must remain explicitly identifiable.

The expanded proposition must not be retrospectively represented as though it had always formed part of the original prediction.

Expanded Boundary $\neq$ Original Prediction

Where expansion materially changes predictive meaning, downstream governance may require reassessment.

21. Boundary Contraction

A Predictive Boundary may also contract.

Removing difficult, failed, unresolved, or contradictory portions of the original predictive scope after predictive performance becomes observable must not silently improve the apparent success of the original prediction.

Contracted Boundary $\neq$ Original Prediction

The original boundary must remain historically recoverable.

22. Retrospective Boundary Drift

PBEM must protect against reinterpretation of predictive boundaries after new information or outcome knowledge becomes available.

A prediction must not be retrospectively widened, narrowed, relocated, reclassified, or otherwise reframed so that an outcome appears to satisfy a proposition that was not originally governed.

Outcome Knowledge $\neq$ Permission to Redefine Predictive Boundaries

This principle supports legitimate later performance evaluation.

23. Boundary Change

Where a material boundary change occurs, PBEM must preserve the distinction between:

Original Boundary

and

Changed Boundary

The change may require reassessment of purpose, target, horizon, methodology, Prediction Basis, assumptions, probability, uncertainty, or feasibility through the applicable PGA™ governance spaces.

PBEM establishes the boundary change.

It does not perform those downstream reassessments.

24. Boundary Continuity

The Predictive Boundary must remain identifiable throughout the Prediction Lifecycle.

Later revisions, supersessions, outcome associations, and performance assessments must be capable of referring to the boundary applicable when the relevant prediction state existed.

Current Predictive Boundary ≠ Complete Boundary History

Historical preservation itself is governed by later PGA™ record and continuity mechanisms.

25. POPS Predictive Frame Completion

PBEM completes the predictive frame established by POPS.

The completed frame consists of:

Prediction Object

Predictive Purpose

Prediction Target

Prediction Horizon

Predictive Boundary

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Governed Prediction Object & Purpose

Only after these elements are sufficiently established should the prediction proceed into methodological governance.

26. Entry Into Prediction Methodology

The completion of POPS does not establish a prediction.

It establishes the governed frame within which a prediction may subsequently be methodologically constructed.

The next governance transition is:

Governed Prediction Object & Purpose → PMES — Prediction Methodology Standard

At that point PGA™ moves from defining what prediction is intended toward governing how that prediction will be methodologically constructed .

Governed Predictive Frame ≠ Governed Prediction

27. Boundary Establishment Sufficiency

A Predictive Boundary is sufficiently established when it:

identifies materially relevant inclusions; identifies materially relevant exclusions; resolves material boundary ambiguity; remains coherent with object, purpose, target, and horizon; distinguishes materially different predictive scopes; identifies relevant dependencies and conditions; can remain historically

distinguishable; supports subsequent methodological governance; prevents silent expansion or contraction of predictive meaning.

Boundary sufficiency does not establish methodological validity, predictive accuracy, probability, feasibility, or correctness.

28. Governance Requirements

PBEM requires that:

1. materially relevant predictive inclusions be identifiable; 2. materially relevant exclusions be identifiable where necessary; 3. relevant boundary dimensions be established; 4. material boundary ambiguity be resolved; 5. boundary specificity remain proportionate to the predictive proposition; 6. the boundary remain coherent with object, purpose, target, and horizon; 7. overlapping predictive scopes remain distinguishable; 8. material boundary dependencies remain identifiable; 9. material boundary changes remain explicit; 10. retrospective boundary drift be prevented; 11. original boundaries remain distinguishable from expanded or contracted boundaries; 12. the Predictive Boundary not be treated as an Operational Reality boundary.

29. Governance Logic

PBEM applies the following operational logic:

Identified Prediction Object

Defined Predictive Purpose

Specified Prediction Target

Defined Prediction Horizon

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Candidate Predictive Boundary

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Relevant Boundary Dimensions Identified

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Inclusions Established

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Exclusions Established where material

↓

Dependencies / Conditions Identified where applicable

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Boundary Ambiguity Assessment

↓

Boundary Coherence Assessment

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Boundary Establishment Sufficiency

↓

Established Predictive Boundary

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Governed Prediction Object & Purpose

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PMES — Prediction Methodology Standard

Where sufficiency is not established:

Boundary Gap → Clarification / Differentiation → Reassessment

30. Minimum Implementation Framework

A minimum implementation of PBEM must establish:

A. Prediction Object Reference

The identified Prediction Object.

B. Predictive Purpose Reference

The defined Predictive Purpose.

C. Prediction Target Reference

The specified Prediction Target.

D. Prediction Horizon Reference

The defined Prediction Horizon.

E. Boundary Statement

A concise representation of the predictive scope.

F. Material Inclusions

Elements explicitly included within the predictive proposition.

G. Material Exclusions

Elements explicitly excluded where necessary to prevent ambiguity.

H. Boundary Dimensions

Applicable geographic, population, system, operational, contextual, conditional, temporal, target, or other dimensions.

I. Boundary Dependencies

External conditions or references upon which the boundary materially depends.

J. Boundary Ambiguity State

Whether unresolved interpretations remain capable of materially changing predictive meaning.

K. Boundary Coherence State

Whether the boundary remains coherent with object, purpose, target, and horizon.

L. Boundary State

At minimum:

Undefined → Candidate → Clarification Required → Established → Changed → Superseded

Only an Established Predictive Boundary completes the POPS predictive frame.

31. Governance Outputs

PBEM may produce:

Established Predictive Boundary; boundary statement; inclusion set; exclusion set; applicable boundary dimensions; boundary dependency references; boundary ambiguity state; boundary coherence state; boundary change state; unresolved boundary requirements; completed POPS predictive frame.

The principal Parent Standard output is:

Governed Prediction Object & Purpose

This becomes the governance input into:

PMES — Prediction Methodology Standard

32. Critical Distinctions

Predictive Boundary $\neq$ Prediction Object

Predictive Boundary $\neq$ Prediction Target

Predictive Boundary $\neq$ Prediction Horizon

Predictive Boundary $\neq$ Operational Reality Boundary

Related Element $\neq$ Included Predictive Element

Boundary Overlap $\neq$ Prediction Equivalence

Boundary Condition $\neq$ Predictive Assumption Governance

Geographic Proximity $\neq$ Predictive Inclusion

Subsystem Prediction $\neq$ Whole-System Prediction

Expanded Boundary $\neq$ Original Prediction

Contracted Boundary $\neq$ Original Prediction

Outcome Knowledge $\neq$ Permission to Redefine Predictive Boundaries

Current Predictive Boundary $\neq$ Complete Boundary History

Governed Predictive Frame $\neq$ Governed Prediction

33. Enterprise Application

PBEM provides organizations, AI systems, forecasting platforms, prediction markets, analytical environments, and autonomous systems with a controlled mechanism for determining exactly where a predictive proposition applies.

This becomes critical where predictions are reused across teams, jurisdictions, populations, systems, markets, or decision environments.

By preserving explicit predictive boundaries, PGA™ can prevent a prediction developed for one governed scope from silently acquiring broader meaning when communicated, automated, aggregated, revised, or evaluated.

34. Use Case 1 — Regional Weather Prediction

A forecasting system predicts severe rainfall during a defined period.

The Prediction Object, Purpose, Target, and Horizon may already be established, but the prediction remains incomplete if its geographic and operational boundaries are unclear.

PBEM establishes whether the prediction applies to a particular city, administrative region, weather zone, infrastructure area, or another defined geographic scope.

Severe rainfall occurring outside that governed boundary cannot retrospectively be used to establish that the original prediction was correct.

35. Use Case 2 — AI System Performance Prediction

An organization predicts that an AI system will achieve a specified operational performance level within a defined horizon.

PBEM establishes whether the prediction applies to a particular model version, deployment environment, task class, user population, infrastructure configuration, or operational context.

Performance achieved by another model version, another environment, or outside the governed conditions must not automatically be attributed to the original predictive proposition.

The boundary therefore preserves the meaning of the prediction before methodology and subsequent predictive governance begin.

36. Canonical Boundary

PBEM governs the establishment, specification, inclusion, exclusion, dimensional structure, coherence, ambiguity, dependency, change, and continuity of the Predictive Boundary that completes the POPS predictive frame. It does not govern predictive methodology, Prediction Basis, assumptions, prediction formation, probability, uncertainty, feasibility, collective prediction, communication, reliance, revision, Operational Reality determination, predictive performance, or Prediction Record continuity.

PBEM establishes where the predictive proposition ends .

It does not establish where Operational Reality ends .

Canonical Closing Statement

A prediction must not become broader, narrower, or different merely because later information makes another interpretation more convenient.

PBEM closes the predictive frame so that the prediction entering methodology remains the same prediction that can later be communicated, revised, compared with reality, and evaluated for performance.

It

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